



DS461: Big Data Analytics

Fall 2026 — GIKI

31 Aug 2026

The Big Question

How do we make data mining and machine learning work when data no longer fits on one computer?

This course answers that question through theory, algorithms, hands-on implementation, and real research.





Why Big Data Matters Now

ML and deep learning have achieved remarkable results — but success hinges on two things: **big data** and **compute**.

Scale Changed Everything - Compute

The algorithms aren't new — what changed is the infrastructure to run them at planetary scale.

More Data = Smarter Models

More data combined with more compute unlocks increasingly capable models and predictions.

THE WORLD OF DATA

NUMBER
OF EMAILS
SENT
EVERY SECOND

2.9

MILLION

DATA
CONSUMED BY
HOUSEHOLDS
EACH DAY

375

MEGABYTES

VIDEO
UPLOADED TO
YOUTUBE EVERY
MINUTE

20

HOURS

DATA PER
DAY
PROCESSED
BY GOOGLE

24

PETABYTES

TWEETS
PER
DAY

50

MILLION

TOTAL MINUTES
SPENT ON
FACEBOOK
EACH MONTH

700

BILLION

DATA SENT
AND RECEIVED
BY MOBILE
INTERNET USERS

1.3

EXABYTES

PRODUCTS
ORDERED ON
AMAZON PER
SECOND

72.9

ITEMS



IN THE 21ST CENTURY, we live a large part of our lives online. Almost everything we do is reduced to bits and sent through cables around the world at light speed. But just how much data are we generating? This is a look at just some of the massive amount of information that human beings create every single day.

<http://spanning.com/blog/choosing-between-storage-based-and-unlimited-storage-for-cloud-data-backup/>

Internet

Billions of Web Pages



www.worldwidewebsize.com www.opte.org

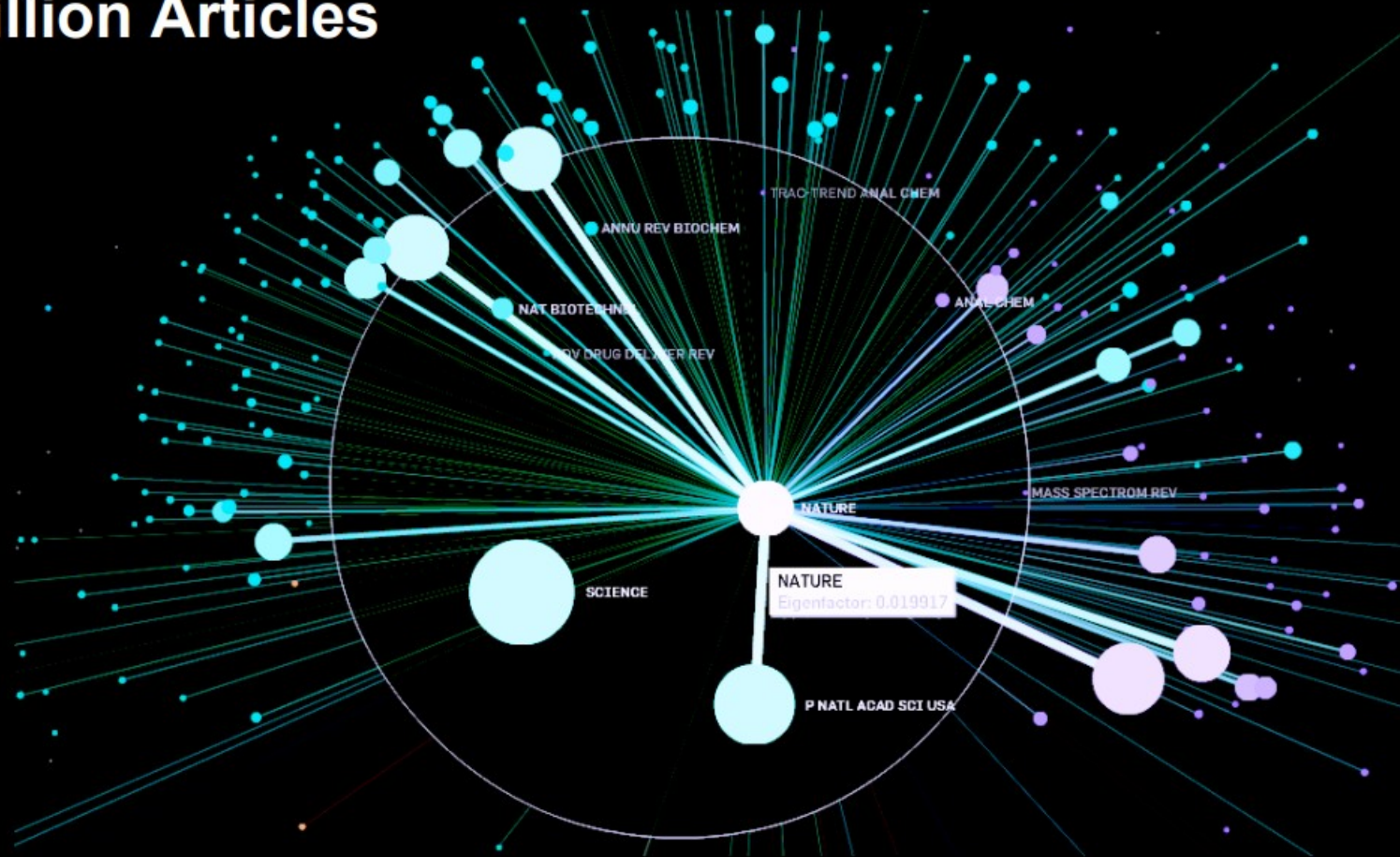
Facebook

3 Billion Users



Citation Network

288+ Million Articles



<https://webofscience.zendesk.com/hc/en-us/articles/48137951657233-Web-of-Science-Coverage-Details> Modified from well-formed.eigenfactor.org

Many More



Who-follows-whom (500+ million users)



Who-buys-what (310+ million users)



AT&T cellphone network

Who-calls-whom (121+ million users)

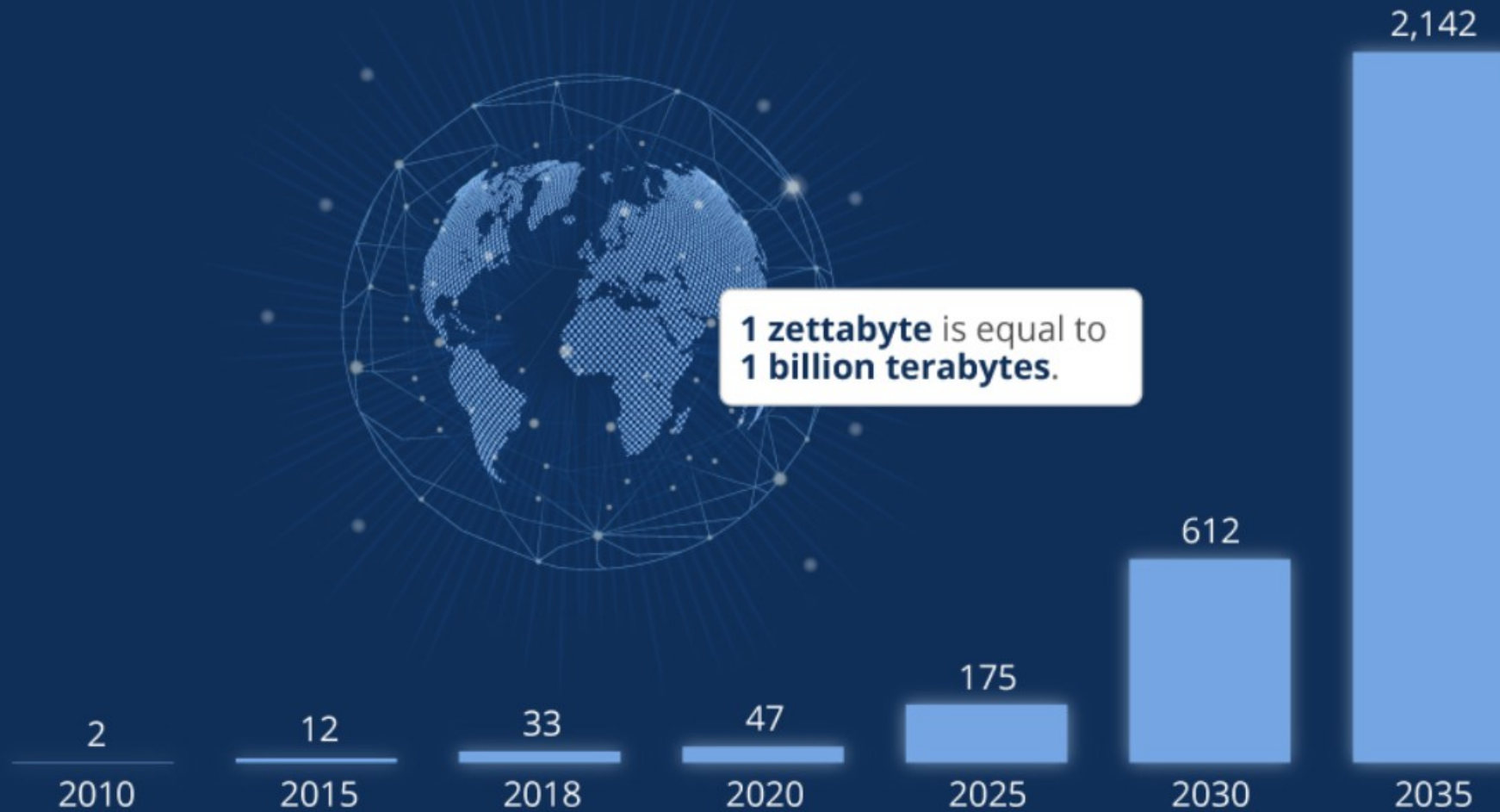
<https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/>

<https://amzscout.net/blog/amazon-statistics/>

https://en.wikipedia.org/wiki/List_of_mobile_network_operators_in_the_United_States

Global Data Creation is About to Explode

Actual and forecast amount of data created worldwide 2010-2035 (in zettabytes)

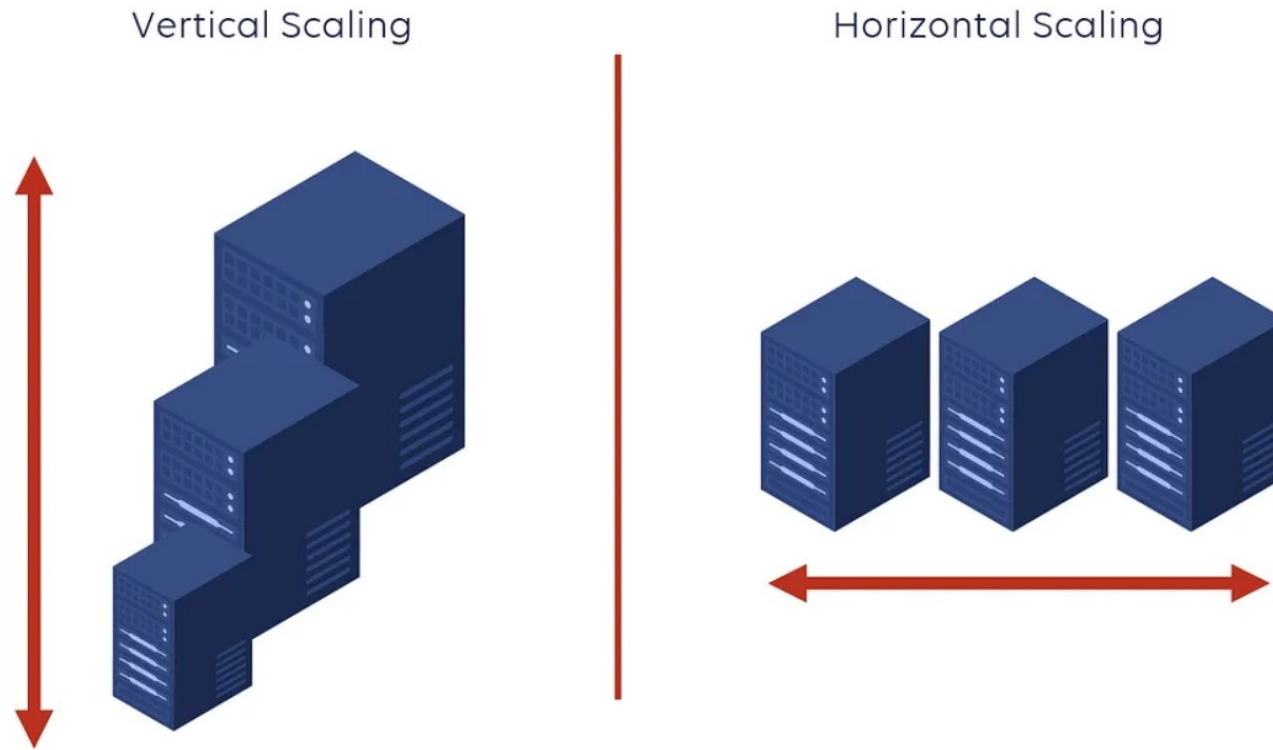


@StatistaCharts

Source: Statista Digital Economy Compass 2019

statista

Distributed Systems, Scalability, and Designing for Failure



Vertical Scaling (Single Machine)

- Add more CPU, memory, disk to one machine.
- Has physical limits.
- If the machine fails → system goes down.
- Low fault tolerance.

Horizontal Scaling (Distributed Systems)

- Add more machines.
- Work is distributed across nodes.
- Leader node assigns tasks to worker nodes.
- If one machine fails → others continue.
- Data is replicated for redundancy

Course at a Glance

3 Credit Hours

15-week semester

Research-Oriented

Theory meets hands-on
implementation

Prerequisites

Linear Algebra, Probability &
Statistics, Programming
Fundamentals

How to Reach Your Instructor



Email

farah.saeed@giki.edu.pk



Office

S18, 2nd Floor, New Academic Block



Office Hours

Tuesday & Thursday, 10:00–11:00
AM – or by appointment

Class Schedule

Day	Time	Room
Monday	4:30–5:20 AM	LH9
Tuesday	8:00–9:00 AM	LH9
Thursday	12:30–1:20 PM	LH9

 All materials, announcements, and resources are shared on **MS Teams**.



COURSE

Contents

Six Modules

01

Foundations

Linear algebra, vectors, matrices, and graphs

03

Similarity & Hashing

Finding similar objects at scale

05

Graph Analytics

PageRank, recommendations, community detection

02

Distributed Frameworks

MapReduce and Apache Spark

04

Dimensionality Reduction

Handling thousands of features efficiently

06

Streams & Scale

Real-time data, large-scale ML, and project presentations

Learning Outcomes

Understand & Analyze

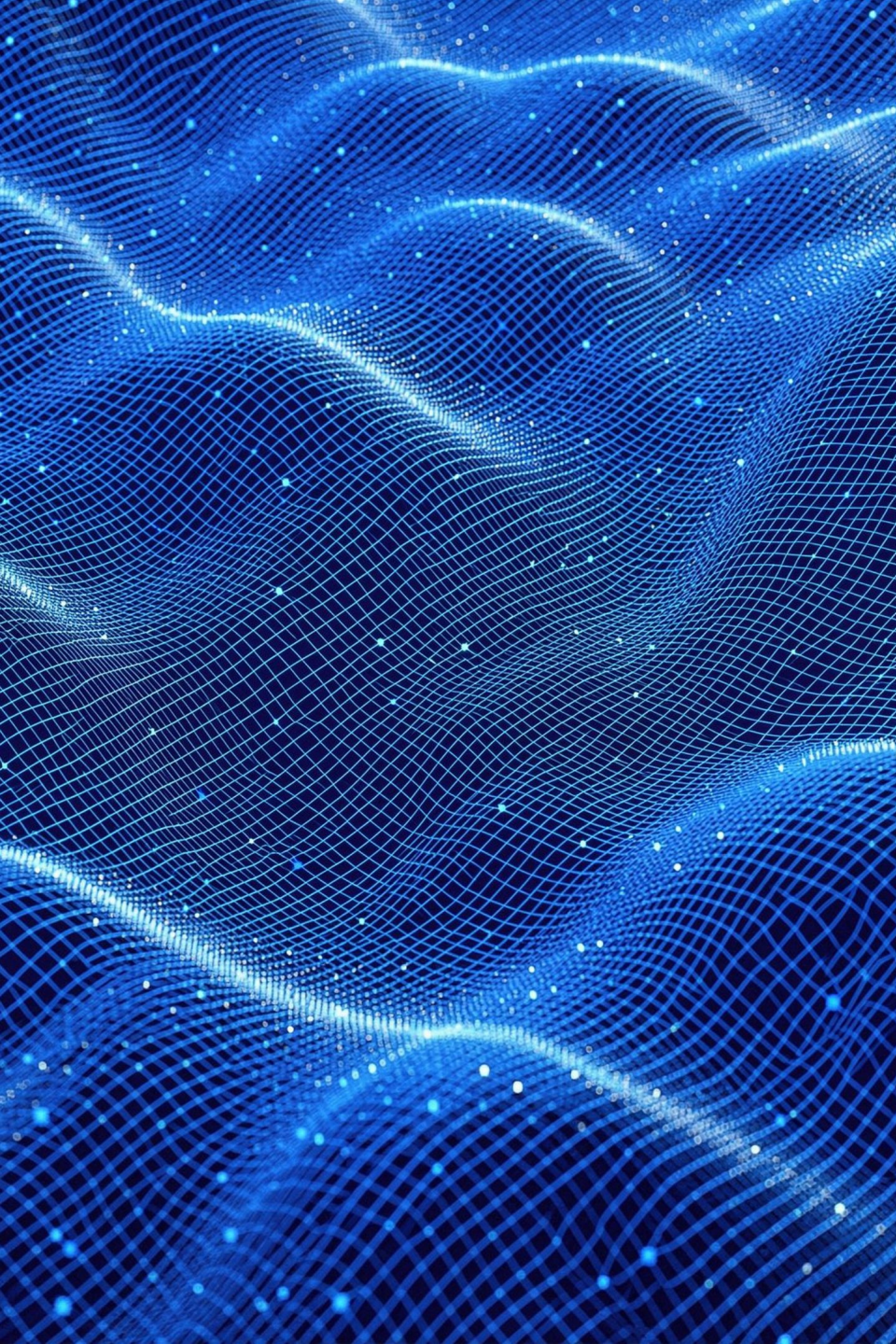
Understand and analyze big data processing frameworks at depth.

Apply at Scale

Apply algorithms for large-scale data mining and machine learning.

Design & Evaluate

Design, implement, and evaluate a research-oriented big-data solution end-to-end.



Your Team Research Project

Team of 4

Each member owns one distinct subproblem

Real-World Dataset

Millions of records tackling a genuine problem

Research Arc

Problem → Literature → Baselines → Method → Experiments → Results

Milestone

Baselines running by mid-semester break

Final Presentation

Improvement from your solution

How You'll Be Assessed

Assignments & Quizzes

5 quizzes + 2 retake quizzes; demos required for all submissions.

Late Work Policy

A grace period may apply, but late submissions may not be accepted after the deadline.

Academic Honesty

Violations result in immediate assessment cancellation — no exceptions.

Attendance

University policy requires 100% attendance throughout the semester.

GRADING

How the final grade breaks down

Weighted across five components, with the two exams carrying most of the grade.



■ Quizzes 10%

■ Assignments 5%

■ Projects 15%

■ Mid Term 30%

■ Final Term 40%

RESOURCES

Text & references

RECOMMENDED TEXT

Mining of Massive Datasets

Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman — 3rd edition

MMDS companion site

mmds.org

Stanford CS246

web.stanford.edu/class/cs246

Stanford CS341

web.stanford.edu/class/cs341

Georgia Tech CSE6242

poloclub.github.io/cse6242